Reconfigurable Series-Parallel Battery Management System

Bridging the 400V to 800V Transition in Electric Vehicles

🚨 The Problem Statement

- **Rigid Architectures:** Current Li-ion battery packs use fixed topologies, limiting adaptability.
- **The 800V Shift:** EVs are moving to 800V architectures for faster charging and lighter wiring.
- **Compatibility Gap:** Adapting 800V EVs to legacy 400V chargers requires complex workarounds (e.g., Porsche Taycan's DC-DC boosters or Macan EV's "bank charging").
- **Motor Trade-offs:** 800V motors offer higher speeds but lower torque than equivalent 400V motors, often requiring heavy multi-speed transmissions (e.g., Porsche Taycan's two-speed gearbox).

💡 The Proposed Solution

A dynamic, low-cost **Reconfigurable Battery Management System (BMS)** that switches topology on-the-fly.

- **Dynamic Switching:** Seamlessly toggles the battery pack between Series (high voltage) and Parallel (high current).
- **Smart Control:** Uses an ESP32 microcontroller for intelligent mode transitions.
- **MOSFET Network:** Employs discrete IRF530/IRF9640 MOSFETs for efficient, converter-less power routing.
- **Real-Time Monitoring:** Continuously tracks cell voltages, current, SOC, and SOH.

📊 Experimental Results

A 3-cell Li-ion prototype was tested under identical charge-discharge conditions.

- **Higher Efficiency:** Series configuration proved significantly more energy-efficient than parallel mode.
- **Energy Savings:** Series mode used **6.7% – 15.3% less DC input** and **12.8% – 16.6% less AC wall energy**.
- **Cost-Effective:** Demonstrates that smart topology reconfiguration is a practical alternative to heavy, expensive DC-DC converters.